

JSON:

JSON (JavaScript Object Notation) is a lightweight data format for storing and exchanging data. It is widely used to send data between a server and a client. JSON is a simple, language-independent, and easy-to-understand format. JSON is "self-describing," making it human-readable.

Syntax: JSON data is represented as key/value pairs, similar to JavaScript objects. Keys are always strings enclosed in double quotes, and values can be strings, numbers, booleans, arrays, objects, or null.

Example: student.json

```
{
  "name": "Shafi",
  "id": "IT21001",
  "age": "23",
  "courses": [
    "E-Commerce and Web Programming",
    "Simulation and Modeling",
    "Cyber Security"
  ],
  "friend": [
    {
      "name": "Meem",
      "id": "IT21059",
      "Dist": "Gazipur"
    },
    {
      "name": "Sagor",
      "id": "IT210",
      "Dist": "Sunamganj"
    }
  ]
}
```

Access data from JSON by JavaScript:

If we want to access "id": "IT21059", then the JS code will be:

(script.js)

```
const data = require("./student.json");

console.log(data.friend[0].id);
```

JavaScript - Web API

A Web API is an application programming interface (API) for the web. The concept of the Web API is not limited to JavaScript only. It can be used with any programming language.

CT What is Web API?

The API is an acronym for the Application Programming Interface. It is a standard protocol or set of rules to communicate between two software components or systems. A web API is an application programming interface for the web.

The API provides an easy syntax to use the complex code. For example, you can use the GeoLocation API to get the coordinates of the users with two lines of code. You don't need to worry about how it works in the backend.

Another real-time example you can take is of a power system at your home. When you plug the cable into the socket, you get electricity. You don't need to worry about how electricity comes into the socket.

There are different types of web APIs, some of which are as follows –

- CT** i. Browser API (Client-Side JavaScript API)
- ii. Server API
- iii. Third Party APIs

i. Browser API (Client-side JavaScript API)

The browser APIs are a set of Web APIs that are provided by the browsers. The browser API is developed on top of core JavaScript, which you can use to manipulate the web page's functionality. There are multiple browser APIs available that can be used to interact with the web page.

Following is a list of common browser APIs –

Storage API – It allows you to store the data in the browser's local storage.

DOM API – It allows you to access DOM elements and manipulate them.

History API – It allows you to get the browser's history.

Fetch API – It allows you to fetch data from web servers.

Forms API – It allows you to validate the form data.

ii. Server API

A server API provides different functionalities to the web server. Server APIs allow developers to interact with the server and access data and resources.

For example, REST API is a server API that allows us to create and consume resources on the server. A JSON API is popular API for accessing data in JSON format. The XML API is a popular API for accessing data in XML format.

iii. **Third-party APIs**

The third-party API allows you to get the data from their web servers. For example, the YouTube API allows you to get the data from YouTube's web server.

Here is the list of common third-party APIs.

YouTube API – It allows you to get YouTube videos and display them on the website.

Facebook API – It allows you to get Facebook posts and display them on the website.

Telegram API – It allows you to fetch and send messages to the Telegram.

Twitter API – It allows you to get tweets from Twitter.

Pinterest API – It allows you to get Pinterest posts.

Fetch API: An Example of Web API- email validation

Documentation: https://docs.abstractapi.com/email-validation?_gl=1*1iymkw2*_gcl_au*MTg4Mjc4MTEyOS4xNzUxMzk3MTA5LjE4MDA5NDk1MTJYuMTc1MTQzMjg3NS4xNzUxNDMyODc1

JS code:

```
const url = 'https://emailvalidation.abstractapi.com/v1/';  
const apiKey = '1f38e6ff3cae4458bf885301abc5a21d';  
const email = 'anowarkabir.mbstu@gmail.com';
```

```
fetch(`${url}?api_key=${apiKey}&email=${email}`)  
  .then(response => response.json())  
  .then(data => {  
    if (data.deliverability === 'DELIVERABLE') {  
      console.log('The email is deliverable.');    }  
    else {  
      console.log('The email is NOT deliverable.');    }  
  })
```